

Preparation for the midterm

There are 10 exercises in this document. The first 5 of them are less complicated than others. The other 5 questions are similar to the examples written in practice classes.

1. Library System

A Library has a collection of books (list of Book objects). Each book has:

- An **Author** (String)
- A **pageCount** (int)
- A **genre** (String)

The librarian needs to perform the following tasks:

1. **Quick Lookup** – Retrieve the first book of a given genre.
2. **Genre Collection** – Return all books in a given genre.
3. **Book Count** – Count the number of books in a given genre.

New books can be added using the **addBook** method.

2. Student Management

A school keeps track of its students. Each **Student** has:

- **Name** (String)
- **Age** (int)
- **Grade** (String, e.g., “10th”, “12th”)

The **School** manages a collection of **Student** objects.

The school should be able to:

1. **Add Student** – Add a new student to the system.
2. **Find Student by Name** – Retrieve a student by their name.
3. **Get Students in a Grade** – Return all students in a given grade.

4. **Count Students in a Grade** – Get the number of students in a particular grade.

3. **Bank Account System**

A bank manages customer accounts. Each **BankAccount** has:

- **Account Number** (String)
- **Owner Name** (String)
- **Balance** (float)

The bank should allow:

1. **Deposit Money** – Add money to an account.
2. **Withdraw Money** – Remove money from an account (if sufficient funds exist).
3. **Check Balance** – Return the balance of an account.

4. **Movie Rental System**

A movie rental shop keeps track of movies. Each **Movie** has:

- **Title** (String)
- **Genre** (String, e.g., “Action”, “Comedy”)
- **Duration** (int, in minutes)

The shop should be able to:

1. **Add Movie** – Add a new movie to the collection.
2. **Find Movies by Genre** – Retrieve all movies of a given genre.
3. **Get Longest Movie** – Find the movie with the longest duration.

5. Ride-Sharing System

A ride-sharing app manages drivers and riders. Each **Driver** has:

- **Name** (String)
- **Car Model** (String)
- **Location** (String)

The system should allow:

1. **Register Driver** – Add a new driver.
2. **Find Driver by Location** – Return all drivers available in a given location.
3. **List All Drivers** – Show all registered drivers.

6. Bank System

A **bank** keeps track of multiple bank accounts. Each **BankAccount** has:

- **Account Number** (String) – A unique identifier for the account.
- **Owner Name** (String) – The name of the account holder.
- **Balance** (float) – The amount of money in the account.

The **Bank** manages a collection of **BankAccount** objects.

The Bank class should provide:

1. **Add Account** – Add a new bank account to the system.
2. **Find Account by Owner Name** – Retrieve an account using the owner's name.
3. **Deposit Money** – Add money to an account by providing the account number and deposit amount.

4. **Withdraw Money** – Withdraw a given amount from an account (if sufficient balance is available).
5. **Check Balance** – Return the balance of a given account using its account number.
6. **Close Account** – Remove an account from the system when it is no longer needed.
7. **Get Total Bank Balance** – Return the total amount of money across all accounts.

7. Movie Rental System

A **movie rental system** keeps track of available movies for rental. Each **Movie** has:

- **Title** (String) – The name of the movie.
- **Genre** (String) – The category of the movie (e.g., “Action”, “Comedy”).
- **Duration** (int) – The runtime of the movie in minutes.
- **Available** (Boolean) – Indicates whether the movie is currently available for rental.

The **MovieRentalSystem** manages a collection of **Movie** objects.

The MovieRentalSystem class should provide:

1. **Add Movie** – Add a new movie to the system.
2. **Find Movies by Genre** – Retrieve all movies that belong to a specific genre.
3. **Get Longest Movie** – Find and return the movie with the longest duration.
4. **Remove Movie** – Delete a movie from the system.
5. **List All Movies** – Return a list of all available movies in the system.
6. **Check Availability** – Check if a specific movie is available for rent.
7. **Rent Movie** – Mark a movie as rented, making it unavailable for others until returned.

8. Online Shopping System

An **online store** sells different products. Each **Product** has:

- **Name** (String) – The name of the product.
- **Price** (float) – The cost of the product.

The **Store** manages a collection of **Product** objects and allows customers to add products to a shopping cart. The cart is an attribute in Store class.

The Store class should provide:

1. **Add Product to Store** – Add a new product to the store's inventory.
2. **Remove Product from Store** – Remove a product from the store.
3. **Find Product by Name** – Retrieve a product from the store by its name.
4. **Add Product to Cart** – Allow a customer to add a product to their cart.
5. **Remove Product from Cart** – Remove a product from the cart before checkout.
6. **View Cart** – Display all products currently in the cart.
7. **Checkout** – Calculate the total cost of all items in the cart.

9. Ride-Sharing System

A **ride-sharing app** manages available drivers for customers looking for a ride. Each **Driver** has:

- **Name** (String) – The name of the driver.
- **Car Model** (String) – The type of car the driver owns.

- **Location** (String) – The city or area where the driver is currently available.
- **Availability** (Boolean) – Indicates whether the driver is currently accepting rides.

The **RideSharingApp** manages a collection of **Driver** objects.

The RideSharingApp class should provide:

1. **Register Driver** – Add a new driver to the system.
2. **Find Drivers by Location** – Retrieve all drivers available in a given location.
3. **List All Drivers** – Return a list of all drivers registered in the system.
4. **Remove Driver** – Remove a driver from the system if they stop working for the app.
5. **Mark Driver Availability** – Update a driver’s availability status (available or unavailable).
6. **Find Available Drivers** – Retrieve a list of all drivers who are currently available for rides.
7. **Request Ride** – Find an available driver in a given location.

10. Hotel Booking System

A **hotel** manages its rooms . Each **Room** has:

- **Room Number** (int) – A unique identifier for each room.
- **Type** (String) – The category of the room (e.g., “Single”, “Double”).
- **Price per Night** (float) – The cost of staying in the room for one night.
- **Available** (Boolean) – Indicates whether the room is currently available for booking.

The **Hotel** manages a collection of **Room** objects.

The Hotel class should provide:

1. **Add Room** – Add a new room to the hotel.

2. **Find Available Rooms by Type** – Retrieve all available rooms of a specific type.
3. **Book Room** – Make the availability of the room object unavailable.
4. **Cancel Booking** – Mark a room as available again.
5. **Get Total Available Rooms** – Return the number of rooms that are available.
6. **List All Rooms** – Display all rooms in the hotel, including booked and available ones.
7. **Find Most Expensive Room** – Return the room with the highest price per night.